

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Application-Based Questions (3 Questions)

Course Code:	ITA0609	Course Title:	Machine Learning
CO Assessed:	CO1 – Analyze (BL 3)	Assessment:	Assessment Tool 1 – Application Based Questions.
Weightage:	40% of CIA	Total Marks:	30 (3 Questions ×10 Mark)
CO1 Statement:	<i>To acquire a foundational understanding of key machine learning concepts including version spaces, candidate elimination, inductive bias, and heuristic search (BL1)</i>		
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Section / Batch:	SLOT A	Date:	14.7.26

Code of Conduct

I (T.Gobika / 192312286) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:T.Gobika

Instructions.

- This test contains 5 Short questions, each carrying 10 mark.
- All questions are application-based and require analytical thinking and practical implementation. · No negative marking. Unanswered questions carry zero marks.
- Provide suitable algorithms, logic, calculations, or case-based solutions wherever required.

Annexure A – Application-Based Questions

1. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?(5 Marks)

A parking facility has three sections containing cars of different colors.

- **Section A:** 5 Red, 3 Blue, 2 Black
- **Section B:** 4 Red, 6 Blue, 5 Black
- **Section C:** 7 Red, 2 Blue, 6 Black

A section is chosen at random, but **Section B is twice as likely to be selected** as each of the other sections.

A randomly selected car from the chosen section is **Blue**.

Find the probability that the car came from **Section C**.

Step 1: Given Data

Section A

- Red = 5
- Blue = 3
- Black = 2

Total cars

$$5 + 3 + 2 = 10$$

Section B

- Red = 4
- Blue = 6
- Black = 5

Total cars

$$4 + 6 + 5 = 15$$

Section C

- Red = 7
- Blue = 2
- Black = 6

Total cars

$$7 + 2 + 6 = 15$$

Step 2: Define Events

Let

- A = Car is selected from Section A
- B = Car is selected from Section B
- C = Car is selected from Section C
- Blue = Selected car is blue

We need to find

$$P(C \mid \text{Blue})$$

Step 3: Prior Probabilities

The question says

Section B is twice as likely to be selected as Section A and Section C.

Assume

Section A = x

Section C = x

Section B = $2x$

Total probability

$$x + 2x + x = 4x = 1$$

Therefore,

$$x = \frac{1}{4}$$

Hence,

$$\begin{aligned} P(A) &= \frac{1}{4} \\ P(B) &= \frac{1}{2} \\ P(C) &= \frac{1}{4} \end{aligned}$$

Step 4: Conditional Probabilities

Probability of selecting a blue car from each section.

Section A

$$P(\text{Blue} \mid A) = \frac{3}{10}$$

Section B

$$P(\text{Blue} \mid B) = \frac{6}{15} = \frac{2}{5}$$

Section C

$$P(\text{Blue} \mid C) = \frac{2}{15}$$

Step 5: Bayes' Theorem

The formula is

$$P(C \mid \text{Blue}) = \frac{P(\text{Blue} \mid C) \times P(C)}{P(\text{Blue})}$$

where

$$P(\text{Blue}) = P(\text{Blue} \mid A)P(A) + P(\text{Blue} \mid B)P(B) + P(\text{Blue} \mid C)P(C)$$

Step 6: Find $P(\text{Blue})$

Substitute the values

$$P(\text{Blue}) = \frac{3}{10} \times \frac{1}{4} + \frac{2}{5} \times \frac{1}{2} + \frac{2}{15} \times \frac{1}{4}$$

Calculate each term

$$= \frac{3}{40} + \frac{1}{5} + \frac{1}{30}$$

Take LCM = 120

$$\begin{aligned} &= \frac{9}{120} + \frac{24}{120} + \frac{4}{120} \\ &= \frac{37}{120} \end{aligned}$$

Therefore,

$$P(\text{Blue}) = \frac{37}{120}$$

Step 7: Apply Bayes' Theorem

$$P(C \mid \text{Blue}) = \frac{\frac{2}{15} \times \frac{1}{4}}{\frac{37}{120}}$$

Simplify numerator

$$= \frac{1}{30}$$

Hence,

$$= \frac{\frac{1}{30}}{\frac{37}{120}}$$

Multiply by reciprocal

$$= \frac{1}{30} \times \frac{120}{37}$$
$$= \frac{4}{37}$$

Step 8: Final Answer

$$P(C \mid \text{Blue}) = \frac{4}{37}$$

Decimal form

$$\frac{4}{37} \approx 0.1081$$

or

$$10.81\%$$

Conclusion

Using **Bayes' Theorem**, the probability that the selected blue car came from **Section C** is

$$\frac{4}{37} \approx 10.81\%$$

2. **Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)**
1. (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
 2. (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
 3. (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
 4. (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
 5. (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
 6. (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
 7. (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
 8. (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2
 9. i) Identify the epoch at which the model begins to overfit.
 10. ii) Suggest two effective techniques to reduce overfitting in this model.
 11. iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Aim:

To analyze the performance of a machine learning model using training and validation loss over multiple epochs, identify the point of overfitting, and suggest techniques to improve the model's generalization performance.

Given Data

Epoch	Training Loss	Validation Loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

Understanding the Data

Training Loss:

Training loss measures how well the model fits the **training dataset**. A lower training loss indicates that the model is learning the training data.

Validation Loss:

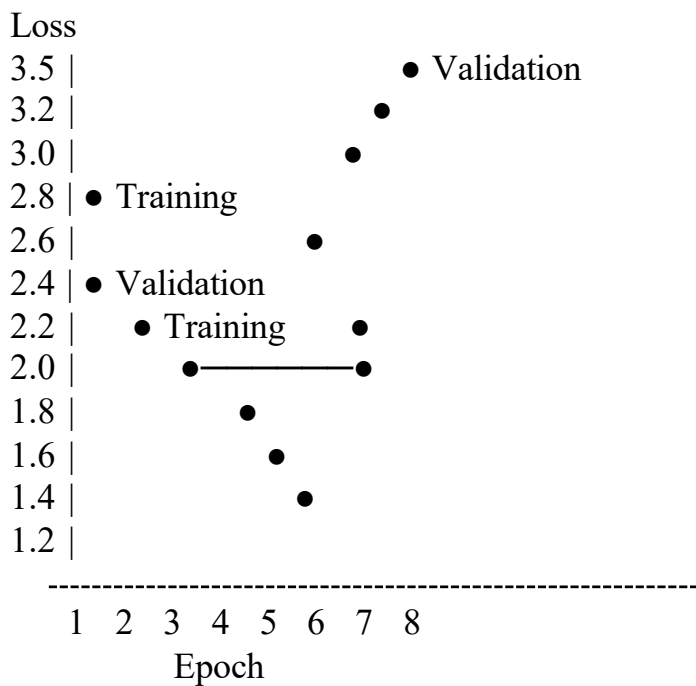
Validation loss measures how well the model performs on **new (unseen) data**. A lower validation loss indicates better generalization.

Observation Table

Epoch	Observation
1	Both losses are high because learning has just started.
2	Training and validation losses decrease, indicating improved learning.
3	Both losses continue to decrease, showing effective learning.
4	Training loss decreases, but validation loss increases slightly. This is the first sign of overfitting.
5	Training loss continues to decrease, while validation loss increases further. Overfitting becomes more evident.
6	Validation loss rises significantly, indicating the model is memorizing the training data.
7	Overfitting increases as the gap between training and validation loss widens.
8	Severe overfitting is observed, with the highest validation loss.

Graph

A line graph helps visualize the trend.



Analysis

Training Loss Trend

- Training loss decreases continuously from **2.8 to 1.3**.
- This indicates the model is learning the training data effectively.

Validation Loss Trend

- Validation loss decreases until **Epoch 3**.
- From **Epoch 4 onward**, validation loss increases while training loss continues to decrease.

This pattern indicates **overfitting**.

(i) Identify the epoch at which the model begins to overfit.

A model begins to overfit when the training loss continues to decrease but the validation loss starts increasing.

From the table:

- Epoch 3: Validation Loss = **2.0**
- Epoch 4: Validation Loss = **2.1** (starts increasing)

Therefore,

The model begins to overfit at Epoch 4.

(ii) Suggest two effective techniques to reduce overfitting.

1. Early Stopping

Training is stopped when the validation loss starts increasing. In this case, training should stop around **Epoch 3 or 4** to prevent overfitting.

2. Regularization (L1 or L2)

Regularization adds a penalty to large model weights, reducing model complexity and helping it generalize better.

(Other valid techniques include Dropout, Data Augmentation, Collecting More Data, or Simplifying the Model.)

(iii) Discuss how overfitting affects the model's generalization ability.

Overfitting causes the model to memorize the training data instead of learning general patterns. As a result:

- Training accuracy becomes very high.
- Validation accuracy decreases.
- The model performs poorly on unseen data.

- Predictions become less reliable in real-world applications.

Thus, overfitting reduces the model's ability to generalize effectively.

Conclusion

The training loss consistently decreases across all epochs, indicating successful learning on the training dataset. However, the validation loss starts increasing after **Epoch 3**, showing that the model begins to overfit from **Epoch 4** onward. Techniques such as **Early Stopping** and **Regularization** can help reduce overfitting and improve the model's performance on unseen data.

3. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

12. (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
13. (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
14. (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
15. (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
16. (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
17. (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
18. (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
19. (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%
20. i) Identify the epoch at which the model starts overfitting.
21. ii) Explain why validation accuracy decreases despite training accuracy increasing.
22. iii) Suggest two methods to improve validation performance.

Performance Analysis of a Machine Learning Model Using Training and Validation Accuracy

Aim

To analyze the performance of a machine learning model using training and validation accuracy over multiple epochs, identify the point of overfitting, and suggest methods to improve the model's validation performance.

Given Data

Epoch Training Accuracy (%) Validation Accuracy (%)

1	60	58
2	65	63
3	70	68
4	75	72
5	80	73
6	85	72

Epoch Training Accuracy (%) Validation Accuracy (%)

7	88	70
8	92	68

Understanding the Data

Training Accuracy

Training accuracy indicates how well the model performs on the **training dataset**. A higher training accuracy means the model is learning the training data effectively.

Validation Accuracy

Validation accuracy measures how well the model performs on **unseen (validation) data**. A higher validation accuracy indicates better generalization.

Observation Table

Epoch Observation

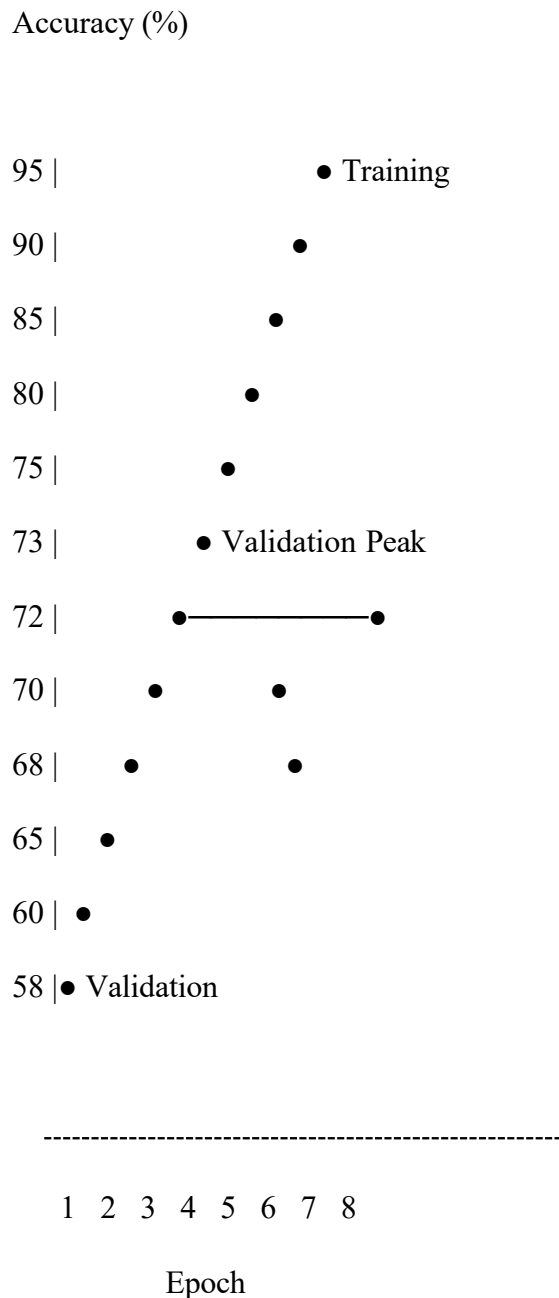
1	Both training and validation accuracy are low because the model has just started learning.
2	Both accuracies increase, indicating effective learning.
3	Training and validation accuracy continue to improve steadily.
4	Both accuracies increase, showing the model is still learning general patterns.
5	Validation accuracy reaches its highest value (73%), while training accuracy continues to increase.
6	Training accuracy increases, but validation accuracy decreases. This is the first indication of overfitting.
7	Training accuracy continues to improve, while validation accuracy decreases further.
8	Training accuracy reaches 92%, but validation accuracy drops to 68%, indicating severe overfitting.

Accuracy Graph

Draw a **line graph** showing both accuracies.

- **X-axis:** Epoch
- **Y-axis:** Accuracy (%)
- **Line 1:** Training Accuracy
- **Line 2:** Validation Accuracy

A simple representation:



Performance Analysis

Training Accuracy Trend

- Training accuracy increases steadily from **60% to 92%**.
- This indicates the model is learning the training data effectively.

Validation Accuracy Trend

- Validation accuracy increases from **58% to 73%** up to **Epoch 5**.
- From **Epoch 6 onward**, validation accuracy decreases even though training accuracy continues to increase.

This behavior indicates that the model is **overfitting**.

(i) Identify the epoch at which the model starts overfitting.

Overfitting begins when:

- Training accuracy keeps increasing.
- Validation accuracy starts decreasing.

From the table:

- Epoch 5: Validation Accuracy = **73%** (highest)
- Epoch 6: Validation Accuracy = **72%** (starts decreasing)

Therefore,

The model starts overfitting at Epoch 6.
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(ii) Explain why validation accuracy decreases despite training accuracy increasing.

As training continues, the model starts memorizing the training data instead of learning general patterns. This causes:

- Higher training accuracy because the model fits the training data very closely.
- Lower validation accuracy because the model cannot perform well on unseen data.

This phenomenon is called **overfitting**, where the model loses its ability to generalize to new data.

(iii) Suggest two methods to improve validation performance.

1. Early Stopping

Stop training when the validation accuracy stops improving or begins to decrease. In this example, training should stop around **Epoch 5** to avoid overfitting.

2. Dropout Regularization

Introduce dropout layers during training. Dropout randomly deactivates some neurons in each training iteration, preventing the model from relying too heavily on specific features and improving its ability to generalize.

Conclusion

The training accuracy increases consistently from **60% to 92%**, indicating that the model learns the training data well. The validation accuracy improves only up to **Epoch 5** and then decreases from **Epoch 6**, even though training accuracy continues to rise. This indicates that the model begins **overfitting at Epoch 6**. Applying techniques such as **Early Stopping** and **Dropout Regularization** can improve validation performance and enhance the model's ability to generalize to unseen data.